

17–(MathAA/10_HL_Winter_2021_Q11)–MathematicalInduction

[Maximum mark: 14]

(a) Prove by mathematical induction that $\frac{d^n}{dx^n}(x^2e^x) = [x^2 + 2nx + n(n-1)]e^x$ for $n \in \mathbb{Z}^+$. [7]

(b) Hence or otherwise, determine the Maclaurin series of $f(x) = x^2e^x$ in ascending powers of x , up to and including the term in x^3 . [3]

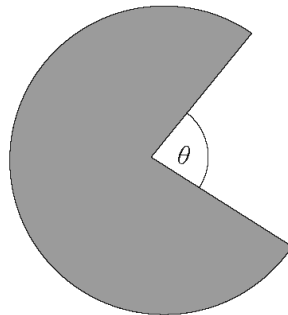
(c) Hence or otherwise, determine the value of $\lim_{x \rightarrow 0} \left[\frac{(x^2e^x - x^2)^{1/3}}{x^9} \right]$. [4]



1-(MAT/11_HL_Summer_2015_Q1)-Radian

The logo, for a company that makes chocolate, is a sector of a circle of radius 2 cm, shown as shaded in the diagram. The area of the logo is $3\pi \text{ cm}^2$.

diagram not to scale



- (a) Find, in radians, the value of the angle θ , as indicated on the diagram. [3]
- (b) Find the total length of the perimeter of the logo. [2]

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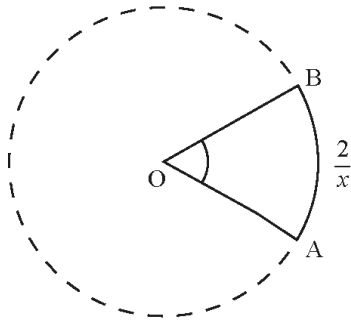


2-(MAT/10_HL_Winter_2015_Q1)-Radian

[Maximum mark: 4]

The following diagram shows a sector of a circle where $\angle AOB = x$ radians and the length of the arc $AB = \frac{2}{x}$ cm.

Given that the area of the sector is 16 cm^2 , find the length of the arc AB .



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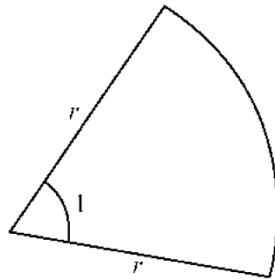
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3-(MAT/11_HL_Summer_2019_Q3)-Radian

[Maximum mark: 4]

A sector of a circle with radius r cm, where $r > 0$, is shown on the following diagram. The sector has an angle of 1 radian at the centre.



Let the area of the sector be A cm² and the perimeter be P cm. Given that $A = P$, find the value of r .

